The massive gravity dam

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Abstract

The gravity dam has played an important role in the environment around us, it solves the four grand challenges that are faced by Egypt. Sources of clean water are increased by it, urban congestion and climate change are decreased by it, and recycled garbage and wastes are used for economic and environmental purposes. Constructing a gravity dam to store rainwater in rocky and desert areas has been a perfect solution to solve the problems of urban congestion by constructing a road (bridge) at the top of the dam so urban congestion and traffic jams are decreased. sources of clean water are increased by storing the water in the reservoirs. Climate change has been reduced by reducing urban congestion and the pollutants in the air, water, and soil. Recycled garbage and waste have been used to construct an eco-friendly dam. Some design requirements must be calculated and considered before constructing the dam, so, care was taken of them to ensure the success of the dam. A gravity dam was built to withstand the pressure of water falling in its body and facing it by its mass. It has been constructed using cement, sand, recycled iron rods, and gallstone. Materials have been chosen according to their resistance to water and hardness. The strength of the dam has been very high, as it can withstand more than 10 kg (about 30 kg) above its body. No deflection or fall occurred. Three test plans have been done, the third one has estimated that the dam takes 25 sec. to pass 14.25 liters (25%) of water.

Introduction

Egypt faces a lot of grand challenges that impede its development and growth. Increasing sources of clean water, dealing with urban congestion, recycling garbage and wastes, and reducing the effect of climate change are determined as some of these grand challenges. So, solving these challenges is our main goal for this semester. First of all, urban congestion is when any area is extremely crowded and has a lot of people or is blocked by traffic. If this problem is solved this will save money to be used in the improvements for other fields and reduce the pollution but, if this problem is not solved climate change, pollution, and ecosystem collapse will increase which will lead to floods occurrence or even drought. So, we built the road (bridge) over the dam to reduce the traffic jams. The second problem is sources of clean water. we are aiming to increase the sources of clean water; we will get the most benefit from torrents with the help of our dam as it will be built between mountains and depressions. So, it will store clean rainfall water, and by that, we will be able to afford a new source of clean water. The third problem is recycling garbage and waste for environmental and economic purposes. Pollution problems have become a series of problems all over the world not only in Egypt, because of getting rid of harmful materials in sources of water, such as metallic cans and plastic. As we have done in our dam, we have used recycled iron rods, recycled wood, recycled plastic, and some eco-friendly materials such as sand and small stones. The fourth and last problem is reducing the effect of climate change. Climate change has increased in the last few years, from the result of climate change is the lack of water in some places because of droughts, so recycling all our wastes in the water coming to the dam will reduce climate change. By all of these, our dam will achieve our main goal by solving all of these problems. We took our experience from the Grande Dixence dam because the Grande Dixence dam (the world's highest gravity dam) was built in Switzerland in 1961. It was constructed to meet Switzerland's rising demand for electricity. The Machu dam failure may be compared with the failure of the Banqiao dam in China in 1975. The Banqiao dam was 25 meters high and 2000 meters long. millions of people were displaced, and tens of thousands of people died directly. The dam was rebuilt in 1991. The advantage of the gravity dam: The gravity dam is a strong dam, provided suitable foundations are available to bear the stresses, and is well adapted for use as an overflow spillway crest. The disadvantages of gravity dams: their initial cost of construction is high, their construction period is comparatively longer, and dams once constructed, cannot be raised further.

Methods

At first, we bought a mold made of recycled wood in the same shape and dimensions as our dam, which is the shape of a gravity dam, and we took many steps to build the dam. We put the gallstone and sweeter cement inside the dam mold, and we worked on leveling the cement inside the mold we bought iron rods 3 mm in diameter and cut them into 40 rods that had 10 cm height to make the dam stronger and more hardness, in addition, we have achieved design requirement. Then, we mixed cement, gallstone, and sand to make the concrete that is going to be poured inside the mold. We poured cement into the mold twice. The first time, we poured half the dam, we waited until it got solid. Then, we poured for the second time. After finishing placing the cement, we waited for it to harden. We waited two days, after that, we separated the dam from the mold and worked on leveling all parts of the dam. We cut part of the tank to connect it to the dam. After that, we connected it to the dam with cement and glue. We placed the gates in the dam and made the test plan that survived in the past. The third time after many attempts

Test plane: Gloves, coats, and goggles for safety have been brought for the test plan. 3 bottles of (20± 0.01) liters of water (PH 1.01) were bought to be poured into the container. The first test plan completely failed. The second test plan almost succeded at first, then failed. The third test plan has completely succeeded and stored (57±0.03) liters of water and (30±2.5) Kg

Materials

Materials	Prices	Photos
Sweeter cement	200 L.E	
Plastic	250 L.E	
Iron rods	30 L.E	
Fix bond	45 L.E	
Tubes	23 L.E	
Gallstone	35 L.E	
Wooden mold	200 L.E	

Total Price: 783 L.E

Results

In the first test of the dam, a lot of water leaked from the container, and it could not hold 7 kg above the dam, the gates were not strong against water, a large part of the dam was destroyed, and many ideas were changed to solve the problems we faced in this attempt. We were also forced to change some resources because water did not leak and exit at the required rate

.The second time, many problems were solved, including the lack of water leaking from the gate, and the dam was able to carry 10 kilograms above the dam, but some errors also occurred, such as the failure of the gates in the dam and the difficulty of controlling the amount of water leaving and entering in precise proportions, which led to difficulty controlling it, but Fewer crashes occurred than in the previous attempt.

the third time it was built a fantastic.

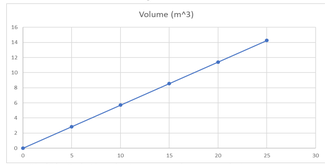
In the third test, the dam was filled successfully, and the tank was filled with water equivalent to 57 liters of water and materials and was used to build the dam. The idea of the dam was very successful, which led to no water leaking from the gate due to the idea and also the method that was carefully chosen and appropriate for the conditions of using the materials. All of this was tried on the test plot, the gates opened and the water came in fine, and the gates closed properly. Airtight and strong to prevent water leakage and excess water from coming out

Test plane 1	Test plane 2	Test plane 3
In this try the dam is not store the 57 L	The dam is store the 57 L	The dam is store 57 L
The dam do not withstand 10 K.G	The dam is withstand more than 10 K.G	The dam is withstand more than 10 K.G
The gate is leaking so much.	The gate is leaking but less than test palne 1	Finely, the gate is not leaking.

Analysis

After finishing our first test plan, we found that our container did not withstand 50 liters of water, it only stored 45 liters then it leaked a lot of water. Therefore, we chose another type of plastic to make our container from it. The body of the dam did not withstand 10 kg on its body once we added 7 kg on its body it fell. So, we chose other materials and learned how to mix them to have the best mixture to form our dam. The flow rate of the gates did not meet our requirements. To solve this problem, we chose better and wider tubes to increase the flow rate. Then we did the second test plan and found that the container withstands (57± 0.03) liters of water (PH 1.01). We found that the dam body was strong and it withstands 10 kg. We found that the gates did not withstand the water pressure and leaked a lot of water. Therefore, we covered the gates with plastic and hung a weight of cement on it. After doing the third test plan We found that the container withstands (57± 0.03) liters (PH 1.01) with no leaking. The dam body withstands (30±2.5) kg (PH 1.01). The gates did not leak, they withstand the water pressure and their flow rate meets our design requirements.

After the success of the third test plan, we started to take some calculations related to the prototype. We calculated the flowrate using the flowrate rule as shown in the figure (ME 1.02):



Fig(2): Relation between volume and time

Q (quantity of flow) = V (volume)/t (time).

14.25 liters (25%)/ 25 sec. = ± 0.57 liters/sec.

We also had to calculate the height of the water inside the container, so, we started our calculations using the physics rule which says that: the volume of water in cm^2/area of the base.

=57,000/ 51*38= (29.4±3) cm.

Table(1): Relation between volume and time

Volume	0	2.85	5.7	8.55	11.4	14.25
Time	0	5	10	15	20	25

From these equations, we have calculated the rate of error in the dam and the height of the water inside the container which is (29.4±3) cm (PH 1.01). These results helped us to determine the correct height of the spillway in the body of the dam. The final results of the test plan made us sure that the dam achieves the design requirements, by withstanding (57±0.03) liters of water (PH 1.01), (30±2.5) Kg (PH 1.01) above the dam's body, the height of stored water is 29.4 cm, the height of the dam is 37 cm, the thickness of the dam in the bottom is 37 cm and in the top is 10 cm, the dam discharge 25%. Finally, our dam has achieved the design requirements and expected results from the first, second, and third test plans. The strong plastic and plastic covers for the gates were great ideas to stop the leaking of water from the dam. There are also positive and negative results that affect the prototype. The positive results are that the flow rate met our expectations and the dam is going to hold greater than the required mass. Secondly, the negative results were the high price of the materials used in the project and we took a lot of time to find the selected material.

Table(2): Learning outcome

subject	L.O	Connection
Physics	1.01	Convert from m to cm using conversion prefix. Calculate accuracy and measurement errors.
Mechanics Computer Science	1.01 1.02	Graphs drawn in the poster. Using Sketch up to create a 3D model of the dam.
Physics	1.03	The mass on the dam's body should by at the center.
Geology	1.03	Determine the building materials we need.
Social Studies	1.02	Determine the constructing place of the dam.
Mathematics	1.04	Determine the ratio of minimization and magnification to have the dam in large scale using similarity.
English		The help uses to write in an academic way.
Mathematics	L.O 2	Using graphs to explain our graphs.

Conclusion

As we finally did the successful test plan, we started taking our observations and results. We found that our prototype stored 57 liters in the container and carried more than 10 kg (about 30 kg). These results are great related to the prototype's cost and failure in the first and second test plan, these results made our prototype achieve the design requirements, which were low price, high efficiency and being eco-friendly. When we compared our project with our friends' projects, we found that our project is completely better than the other projects. For instance, our project has lower price than many other projects with the same shape. Our results provided that our project is much stronger and efficient than any other project. Our solution solves many Egypt grand challenges such as urban congestion and traffic jam, while other projects do not solve this problem. We depend on recycled materials, while other projects depend on industrial materials which increases the pollution. Our dam can hold more than 10 Kg, which shows how strong is it. From all the previous, we can see that our prototype achieves the design requirements in a great way. Finally, our solution is one of the most effective project in the field of flood discharge.

Recommendations

As we all know, nothing is perfect in this world. Being high school students, we did not have full access to resources to make a perfect project, even though we knew how to make a good one. However, we identified some points that could have made the project better if implemented:

1. Build reservoirs in the dam in order to store water at least 25% from the total amount of water.
2. If we applied this dam in the rear life, we should use electric turbines(generators) to get the most benefit of the water flow and generate hydroelectric energy, this thing will help environment.
3. Put some generators in the upper part of the dam but under the road(bridge) in order to exploit the frequent motion of the cars on the bridge to generate electricity which provide the dam with another source of energy.
4. Determine the location of the dam construction, for instance, Sinai.
5. Use a cleric to build the container, this material is expensive put it is so useful and effective.
6. If we applied this dam in the real life, we could build this dam in a desert place to decrease the desertification. Because we use the stored water in the agricultural field.

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